

8 Quince are fruit that grow on trees.

When a quince falls from the tree, two processes take place: browning and decomposition.

- (a) Browning takes place when the fruit is cut and exposed to the air. The browning is catalysed by the enzyme polyphenol oxidase (PPO).

The photographs show browning in a quince cut in half.

Freshly-cut quince
showing some browning



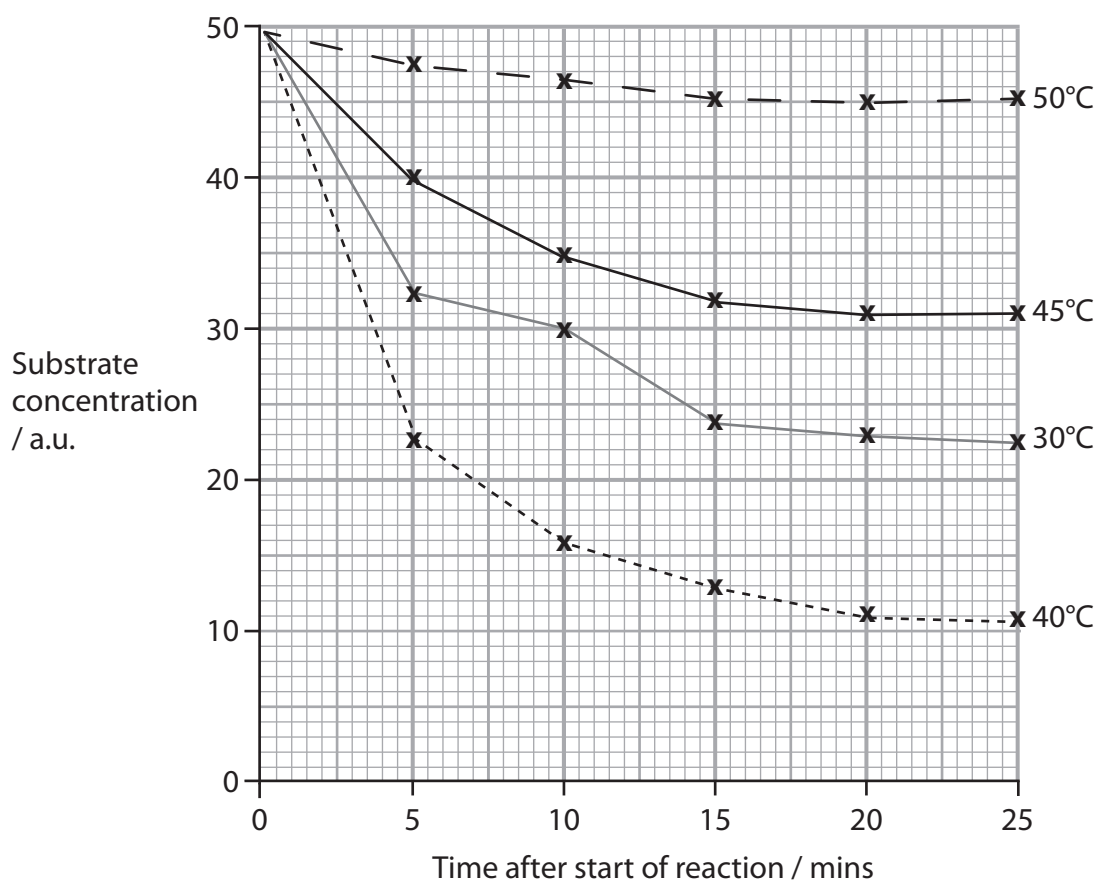
Cut quince exposed to the air
showing extensive browning



(Source: John Adds)

Activity of the enzyme PPO at different temperatures was investigated.

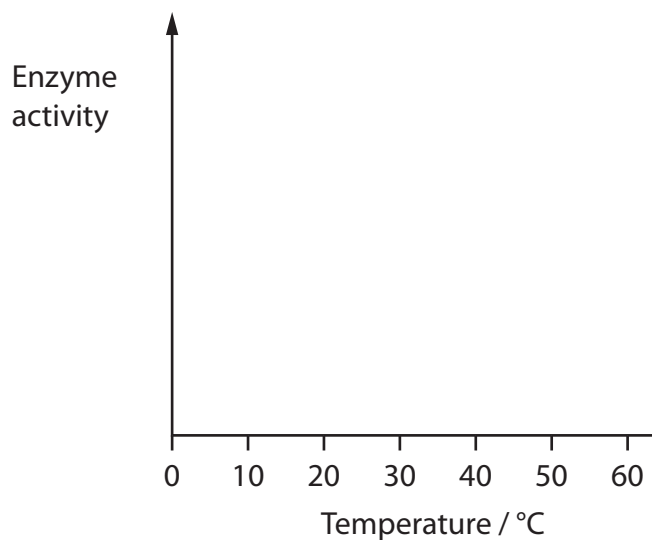
The graph shows the results of this investigation.



- (i) Sketch the graph for enzyme activity plotted against temperature for this enzyme.

Use the data from this investigation.

(2)



- (ii) Calculate the Q_{10} for this enzyme, using the data from this investigation, shown in the first graph.

Use the formula

$$Q_{10} = \frac{R_{t+10}}{R_t}$$

where R_t is the initial rate of reaction at t °C

and R_{t+10} is the initial rate of reaction at $t + 10$ °C.

(4)

Answer.....

(b) A quince fruit is made up of cells and contains a lot of juice.

The table shows the composition of carbohydrates in the juice of one species of quince.

Type of carbohydrate	Name of carbohydrate	Concentration of carbohydrate / mg per 100 cm ³ juice
monosaccharide	fructose	817.2
	glucose	308.3
	inositol	8.3
	rhamnose	12.4
	sorbitol	121.1
	xylose	94.0
disaccharide	sucrose	57.0
trisaccharide	raffinose	8.3
tetrasaccharide	stachyose	9.5

- (i) Complete the table to show the ratio of the concentrations of the four types of carbohydrate.

(2)

Type of carbohydrate	Ratio
monosaccharide	
disaccharide	
trisaccharide	
tetrasaccharide	

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(Total for Question 8 = 15 marks)

TOTAL FOR PAPER = 90 MARKS